

단계1: Dockerfile 확인

docker > samples > 3. Apache 이미지 생성 > Dockerfile > ...

```
1  # Base Image
2  FROM ubuntu:18.04
3
4  # 이미지에 대한 정보
5  LABEL maintainer="my name <my email@gmail.com>"
6
7  # Install Apache
8  RUN apt-get update && \
9      apt-get install -y apache2
10
11 # index.html 소스 수정
12 RUN echo "This is my Apache Web Page" > /var/www/html/index.html
13
14 # 외부에 공개할 port 정의
15 EXPOSE 80
16
17 # Apache 실행
18 CMD [ "/usr/sbin/apache2ctl", "-DFOREGROUND" ]
19
```

단계2: Dockerfile이 있는 폴더로 이동

```
ls
```



A terminal window with a dark background. The top status bar shows a battery icon at 14%, a network icon, and 8.4 GB of memory. The prompt is 'good593' followed by a yellow penguin icon and a blue path '~ / dev / github / course_dev_basic / docker / samples / 3. Apache 이미지 생'. Below the prompt, there is a yellow bar with '성' and a blue bar with 'dev ±'. The command 'ls' is entered in green. The output shows 'Dockerfile', 'README.md', and 'image.png' in white text. Below the output, the prompt and status bar are repeated.

```
good593 ~ / dev / github / course_dev_basic / docker / samples / 3. Apache 이미지 생
성 dev ±
ls
Dockerfile README.md image.png
good593 ~ / dev / github / course_dev_basic / docker / samples / 3. Apache 이미지 생
성 dev ±
```

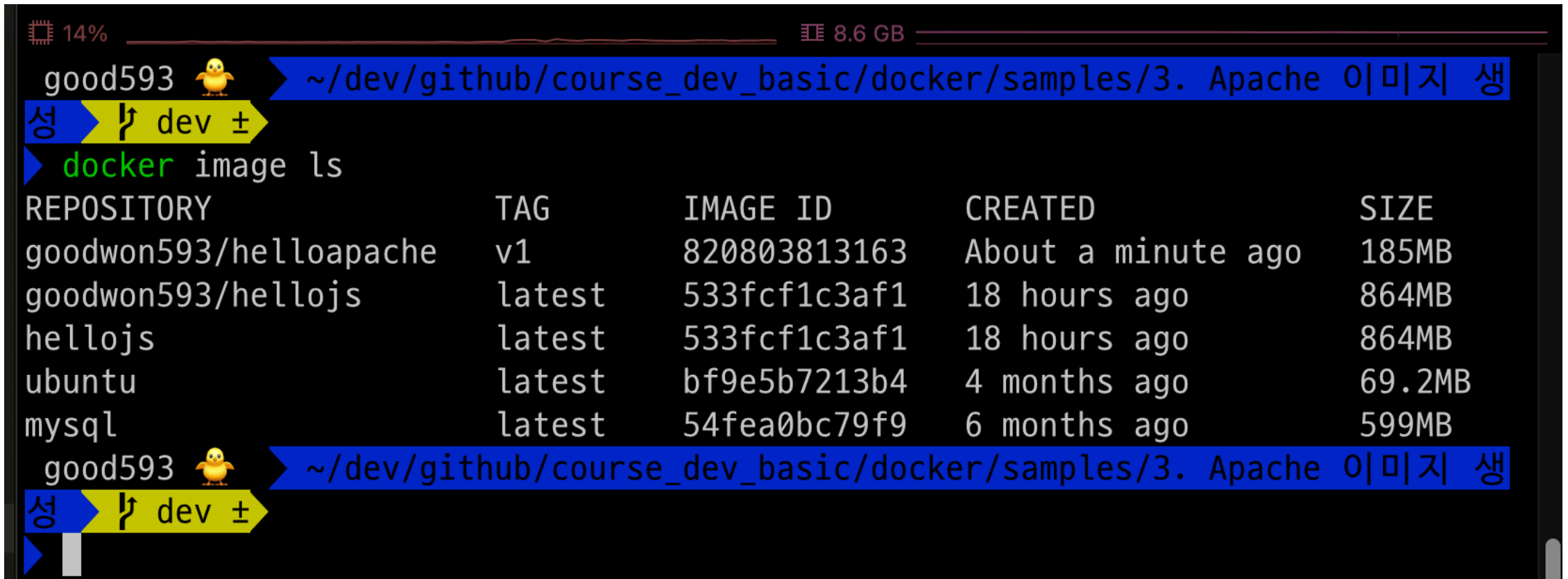
단계3: Dockerfile을 이용하여 이미지 생성

```
docker build -t goodwon593/helloapache:v1 .
```

```
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성
> docker build -t goodwon593/helloapache:v1 .
[+] Building 3.0s (8/8) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 434B 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load metadata for docker.io/library/ubuntu:18.04 2.7s
=> [auth] library/ubuntu:pull token for registry-1.docker.io 0.0s
=> [1/3] FROM docker.io/library/ubuntu:18.04@sha256:152dc042452c496007f0 0.0s
=> CACHED [2/3] RUN apt-get update && apt-get install -y apache2 0.0s
=> [3/3] RUN echo "This is my Apache Web Page" > /var/www/html/index.htm 0.2s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:820803813163658ba5e5b147d75ce3443e69ba58f8b63 0.0s
=> => naming to docker.io/goodwon593/helloapache:v1 0.0s
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성
```

단계4: 생성된 이미지 확인

```
docker image ls
```



The screenshot shows a terminal window with a dark background. At the top, there's a status bar with a battery icon at 14%, a memory icon at 8.6 GB, and a network icon. Below the status bar, the user 'good593' is shown with a yellow penguin icon. The prompt is '~/.dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성'. The user enters 'docker image ls'. The output is a table with 5 columns: REPOSITORY, TAG, IMAGE ID, CREATED, and SIZE. The table lists several images: 'goodwon593/helloapache' (v1, 820803813163, About a minute ago, 185MB), 'goodwon593/hellojs' (latest, 533fcf1c3af1, 18 hours ago, 864MB), 'hellojs' (latest, 533fcf1c3af1, 18 hours ago, 864MB), 'ubuntu' (latest, bf9e5b7213b4, 4 months ago, 69.2MB), and 'mysql' (latest, 54fea0bc79f9, 6 months ago, 599MB). Below the table, the user 'good593' is shown again with the same prompt and a new line for input.

```
good593 ~/.dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성
> docker image ls
REPOSITORY          TAG          IMAGE ID        CREATED         SIZE
goodwon593/helloapache v1           820803813163    About a minute ago 185MB
goodwon593/hellojs   latest       533fcf1c3af1    18 hours ago     864MB
hellojs              latest       533fcf1c3af1    18 hours ago     864MB
ubuntu               latest       bf9e5b7213b4    4 months ago     69.2MB
mysql                latest       54fea0bc79f9    6 months ago     599MB
good593 ~/.dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성
>
```

단계5: 컨테이너 생성

```
docker run --name helloapache-web -d -p 80:80 goodwon593/helloapache:v1
```



A terminal window with a dark background. At the top, there is a status bar showing a battery icon at 18%, a memory icon at 8.8 GB, and a network icon. The terminal shows a user 'good593' with a penguin icon at the prompt, followed by a blue highlighted path '~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생'. Below this, a yellow highlighted prompt '성' is followed by a blue arrow pointing to 'dev ±'. The command 'docker run --name helloapache-web -d -p 80:80 goodwon593/helloapache:v1' is entered, followed by the container ID 'fc232dd8a79cc2375d22f588975f9b90ee7b78ffe2c48fb9c34bb80eae07ab64'. Another instance of the user prompt and path is shown below, followed by another yellow highlighted prompt and blue arrow.

```
good593 ~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 dev ±
> docker run --name helloapache-web -d -p 80:80 goodwon593/helloapache:v1
fc232dd8a79cc2375d22f588975f9b90ee7b78ffe2c48fb9c34bb80eae07ab64
good593 ~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 dev ±
```

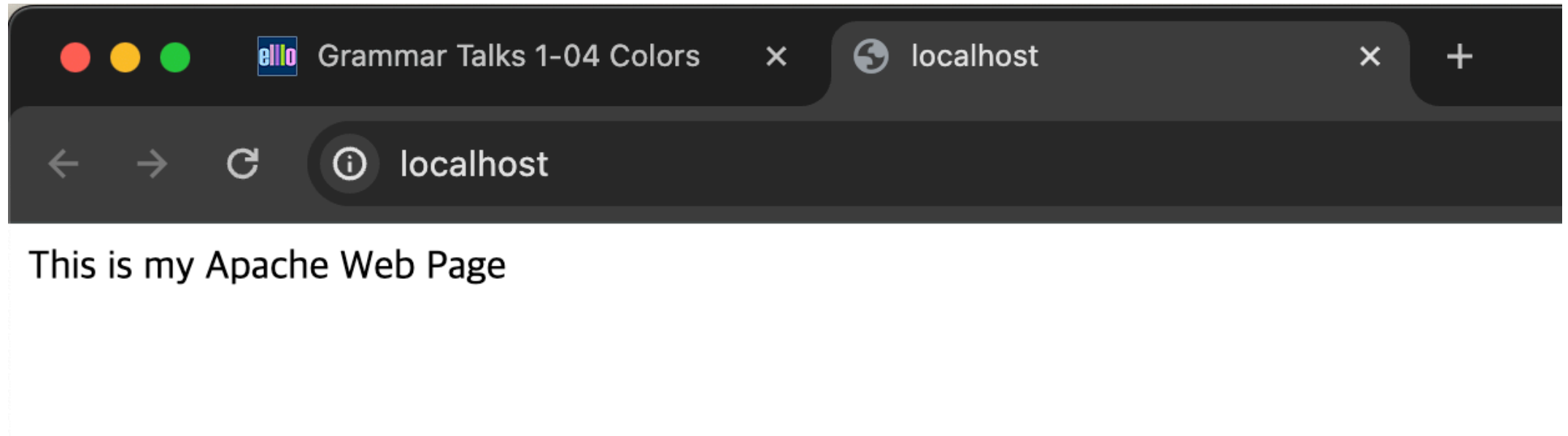
단계6: 작동중인 컨테이너 확인

```
docker ps
```

```
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성
> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED
STATUS        PORTS                               NAMES
fc232dd8a79c   goodwon593/helloapache:v1          "/usr/sbin/apache2ct... 28 seconds a
go Up 27 seconds 0.0.0.0:80->80/tcp        helloapache-web
fd6f8e571db3   mysql                              "docker-entrypoint.s... 3 months ago
Up 23 hours    0.0.0.0:3306->3306/tcp, 33060/tcp  docker-db-1
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성
>
```

단계7: Apache 웹 접속

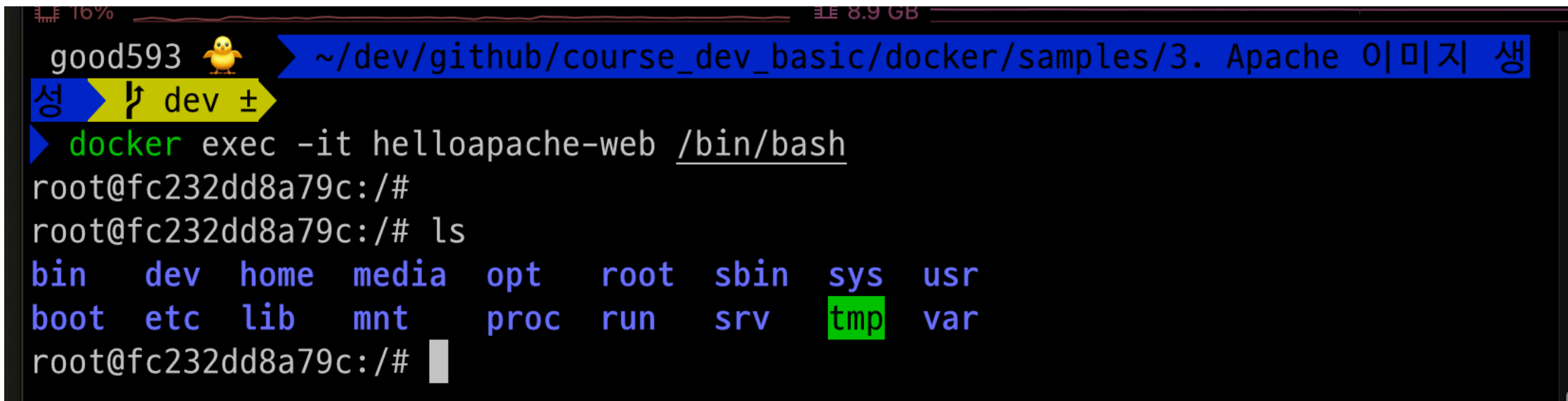
- <http://localhost:80/>



단계8: Apache 컨테이너 접속

- 명령어: `docker exec -it <컨테이너이름> /bin/bash`

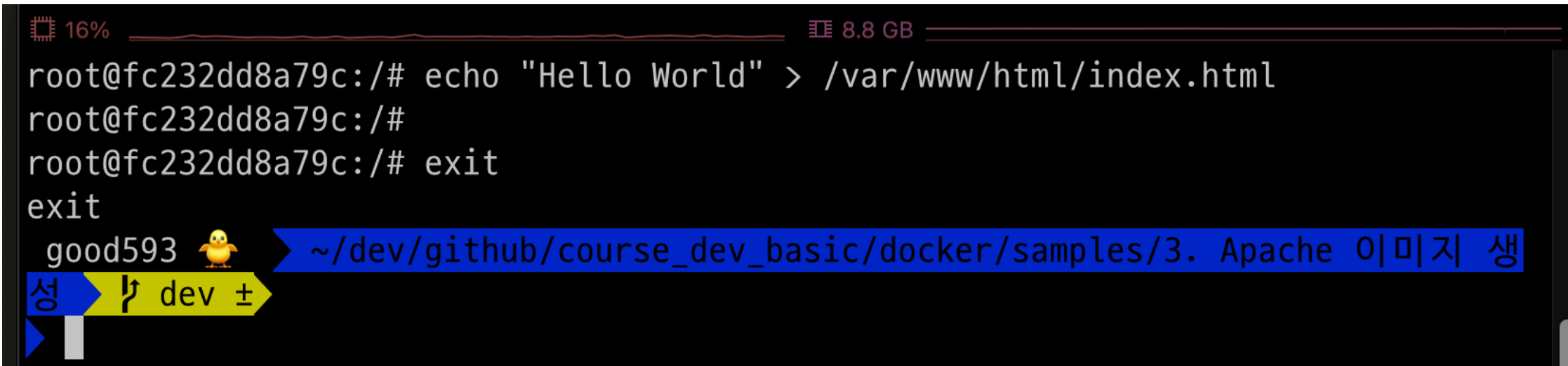
```
docker exec -it helloapache-web /bin/bash
```



```
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성
> docker exec -it helloapache-web /bin/bash
root@fc232dd8a79c:/#
root@fc232dd8a79c:/# ls
bin  dev  home  media  opt  root  sbin  sys  usr
boot  etc  lib  mnt  proc  run  srv  tmp  var
root@fc232dd8a79c:/#
```

단계9: index.html 수정

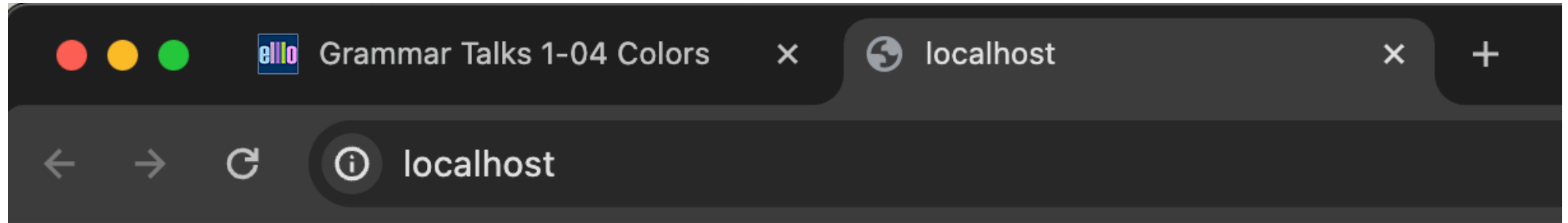
```
echo "Hello World" > /var/www/html/index.html  
exit
```



```
16% 8.8 GB  
root@fc232dd8a79c:/# echo "Hello World" > /var/www/html/index.html  
root@fc232dd8a79c:/#  
root@fc232dd8a79c:/# exit  
exit  
good593 🐧 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생  
성 ↳ dev ±
```

단계10: Apache 컨테이너 접속 > 수정 내용 확인

- <http://localhost:80/>



Hello World

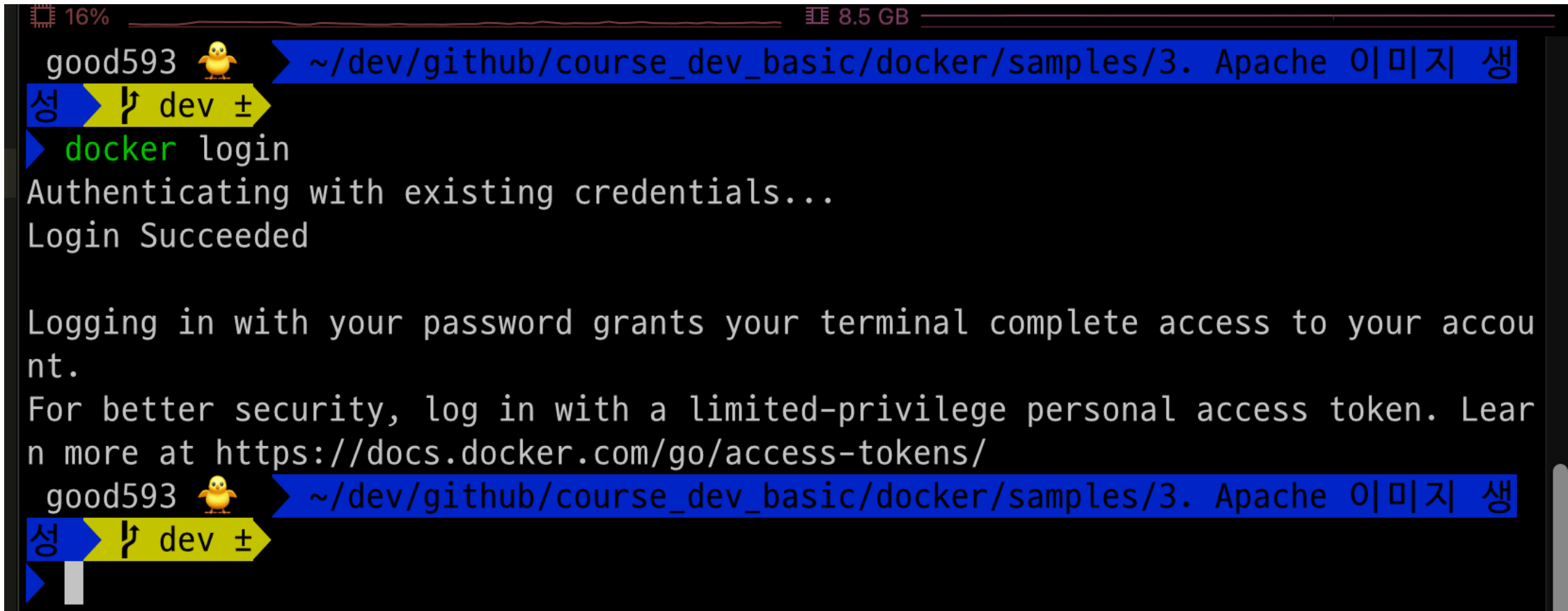
단계11: 실행중인 컨테이너 삭제

```
docker rm -f helloapache-web
docker ps -a
```

```
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 > ? dev ±
> docker rm -f helloapache-web
helloapache-web
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 > ? dev ±
> docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS
PORTS         NAMES
52bb2eb8d414   ubuntu    "/bin/bash"             3 months ago  Exited (129) 3
months ago    ubuntu_test
fd6f8e571db3   mysql     "docker-entrypoint.s..." 3 months ago  Up 24 hours
0.0.0.0:3306->3306/tcp, 33060/tcp  docker-db-1
```

단계12: 도커 로그인

```
docker login
```

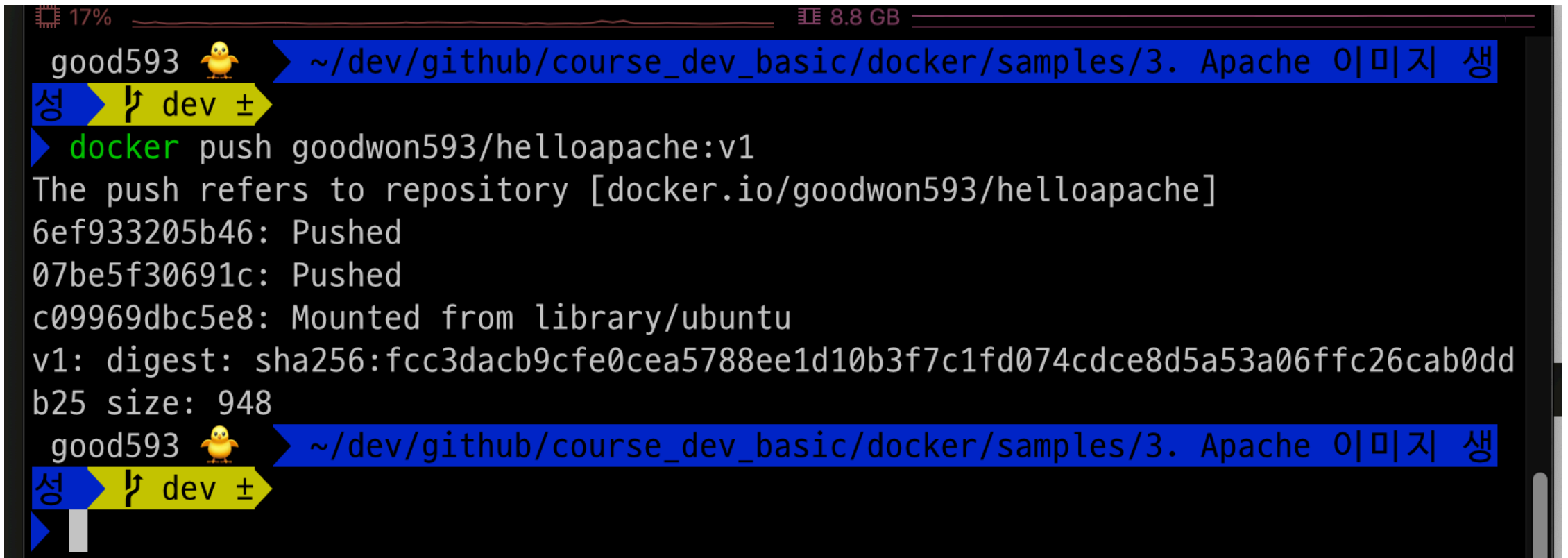
A terminal window with a dark background. At the top, there's a status bar showing '16%' on the left and '8.5 GB' on the right. The terminal prompt is 'good593' followed by a yellow penguin icon and a blue highlighted path '~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성'. Below the prompt, there's a yellow highlighted prompt 'dev ±'. The user enters 'docker login' in green. The terminal shows 'Authenticating with existing credentials...' and 'Login Succeeded'. A message follows: 'Logging in with your password grants your terminal complete access to your account. For better security, log in with a limited-privilege personal access token. Learn more at https://docs.docker.com/go/access-tokens/'. The prompt and path are repeated at the bottom of the terminal.

```
16% 8.5 GB
good593 ~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성
dev ±
> docker login
Authenticating with existing credentials...
Login Succeeded

Logging in with your password grants your terminal complete access to your account.
For better security, log in with a limited-privilege personal access token. Learn
more at https://docs.docker.com/go/access-tokens/
good593 ~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성
dev ±
>
```

단계13: 도커허브로 이미지 업로드

```
docker push goodwon593/helloapache:v1
```



A terminal window with a dark background. The top status bar shows '17%' battery, a network icon, and '8.8 GB' memory. The prompt is 'good593' followed by a penguin icon and a blue highlighted path '~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성'. Below the prompt, a yellow highlighted box contains 'dev ±'. The command 'docker push goodwon593/helloapache:v1' is entered. The output shows the repository path, three layers being pushed (6ef933205b46, 07be5f30691c, c09969dbc5e8), and the final digest and size for the v1 tag. A second identical prompt and yellow box are visible at the bottom.

```
good593 ~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성
dev ±
> docker push goodwon593/helloapache:v1
The push refers to repository [docker.io/goodwon593/helloapache]
6ef933205b46: Pushed
07be5f30691c: Pushed
c09969dbc5e8: Mounted from library/ubuntu
v1: digest: sha256:fcc3dacb9cfe0cea5788ee1d10b3f7c1fd074cdce8d5a53a06ffc26cab0ddb25 size: 948
good593 ~ /dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생성
dev ±
```

단계14: 도커허브에서 이미지 업로드 확인

 Explore Repositories Organizations

Search Docker Hub

+K ? G

goodwon593 Search by repository name All Content Create repository

goodwon593 / helloapache Contains: Image Last pushed: 1 minute ago	Inactive	☆ 0	↓ 0	Public
goodwon593 / hellojs Contains: Image Last pushed: about 17 hours ago	Inactive	☆ 0	↓ 2	Public

단계15: 생성한 이미지 삭제

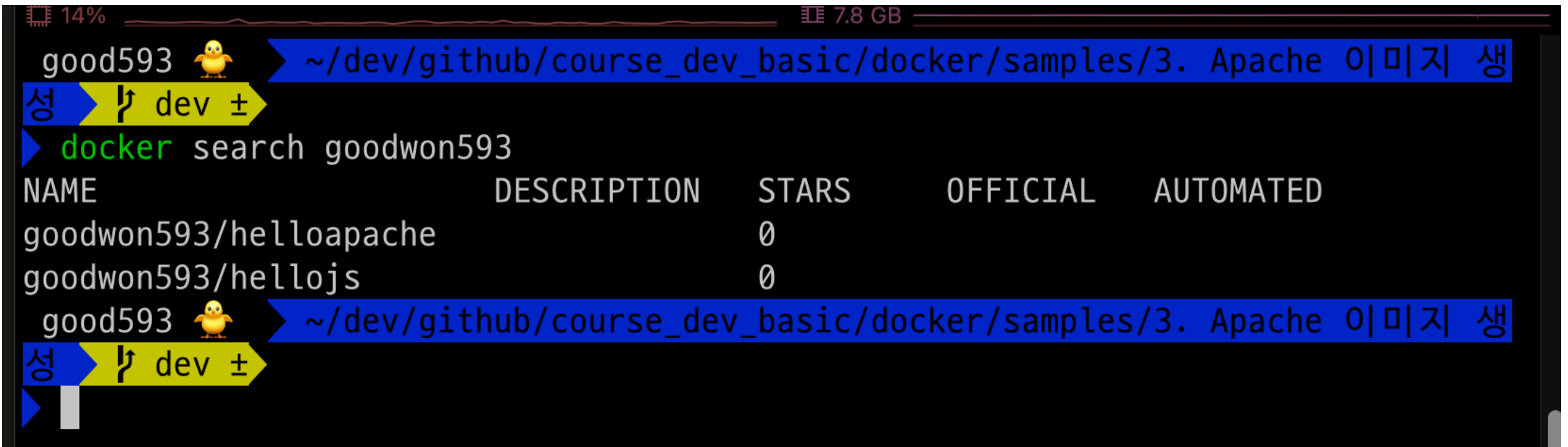
```
docker rmi goodwon593/helloapache:v1
docker image ls
```

```
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 > dev ±
> docker rmi goodwon593/helloapache:v1
Untagged: goodwon593/helloapache:v1
Untagged: goodwon593/helloapache@sha256:fcc3dacb9cfe0cea5788ee1d10b3f7c1fd074cdc
e8d5a53a06ffc26cab0ddb25
Deleted: sha256:820803813163658ba5e5b147d75ce3443e69ba58f8b630e5075bbfcea1a8c0a6
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 > dev ±
> docker image ls
REPOSITORY          TAG          IMAGE ID      CREATED      SIZE
goodwon593/hellojs   latest       533fcf1c3af1  19 hours ago 864MB
hellojs             latest       533fcf1c3af1  19 hours ago 864MB
ubuntu              latest       bf9e5b7213b4  4 months ago 69.2MB
mysql               latest       54fea0bc79f9  6 months ago 599MB
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 > dev ±
```

단계16: 도커허브에서 나의 이미지 조회

- 명령어: `docker search <도커허브아이디>`

```
docker search goodwon593
```



The screenshot shows a terminal window with a dark background. At the top, there's a status bar with '14%' on the left and '7.8 GB' on the right. The prompt is 'good593' followed by a penguin icon and a blue banner with Korean text. Below the prompt, there's a yellow banner with 'dev' and a terminal icon. The command 'docker search goodwon593' is entered. The output is a table with columns: NAME, DESCRIPTION, STARS, OFFICIAL, and AUTOMATED. Two results are shown: 'goodwon593/helloapache' and 'goodwon593/hellojs', both with 0 stars. Below the table, the prompt and banners are repeated.

```
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 ▶ dev ±
> docker search goodwon593
NAME                DESCRIPTION          STARS    OFFICIAL    AUTOMATED
goodwon593/helloapache
goodwon593/hellojs
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 ▶ dev ±
```

단계17: 도커허브에 있는 나의 이미지 다운로드

```
docker pull goodwon593/helloapache:v1
docker image ls
```

```
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 dev ±
> docker pull goodwon593/helloapache:v1
v1: Pulling from goodwon593/helloapache
064a9bb4736d: Already exists
3dbac36e073f: Already exists
d6eed91ec90b: Already exists
Digest: sha256:fcc3dadb9cfe0cea5788ee1d10b3f7c1fd074cdce8d5a53a06ffc26cab0ddb25
Status: Downloaded newer image for goodwon593/helloapache:v1
docker.io/goodwon593/helloapache:v1
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 dev ±
> docker image ls
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
goodwon593/helloapache	v1	820803813163	About an hour ago	185MB
hellojs	latest	533fcf1c3af1	19 hours ago	864MB
goodwon593/hellojs	latest	533fcf1c3af1	19 hours ago	864MB
ubuntu	latest	bf9e5b7213b4	4 months ago	69.2MB
mysql	latest	54fea0bc79f9	6 months ago	599MB

```
good593 ~/dev/github/course_dev_basic/docker/samples/3. Apache 이미지 생
성 dev ±
```